



Airtel: Pricing in the Cannibalisation Era and Transition to Data

On October 1, 2017, Gopal Vittal, Managing Director and Chief Executive Officer-India and South Asia, Bharti Airtel, was in his New Delhi office reviewing current trends and Airtel's position in Indian Telecom. His primary concern was the shifting data consumption trend in the Indian Telecom Industry (Exhibit 1) and the disruptive changes that were impacting pricing and profitability since the entry of Reliance Jio Infocomm Limited (Jio) in September 2016. Data consumption in Indian telecom had started increasing exponentially after the entry of Jio who offered *lifetime free voice services* followed by rock-bottom data tariffs. As Vittal reviewed the data, he wondered if the voice market through a non-VOIP provision was now saturated and would rapidly decline. He was also concerned about the price and revenue implications for Airtel. How might the voice market evolve? *How should he act on the pricing front to enable Airtel revenues to continue to grow in the context of what appeared to be predatory pricing by Reliance Jio?*

Indian Telecom Industry-Current Scenario

The telecom industry in India was the second largest in the world (1206.71 million) in terms of subscribers, with the market split into three segments: Mobile (Wireless), Fixed Line (Wireline) and Internet services. As of September 2017, India had 1206.71 million mobile phone network subscribers and 429.23 million internet subscribers. The nation was the second highest globally for the number of internet users. Mobile data subscriber penetration was at 40% and was expected to double to 80% by 2022.

India had moved from 150 to first position in mobile data usage worldwide. Mobile data consumption in India had surpassed the combined mobile data consumption of the USA and China. The voice to data ratio in India was 75:25 and was changing rapidly (Exhibit 2). Data revenue of the Indian Telecom Industry for the quarter Q3 ending September 2017 rose by 59% to INR 11,900 crore compared to the previous quarter. On a year-on-year basis, data revenue grew double-digits to 27% from INR 9400 crore in Q3 September 2016.¹ The key component of Indian internet usage was mobile-based internet with every 7 of 8 users browsing through their mobile phone.² In India, 300 million mobile phone users had smart phones³; of which 150

¹ Retrieved from: <https://www.financialexpress.com/industry/data-revenue-of-telecom-firms-jumps-by-staggering-59-pct-to-rs-11900-cr-in-q2/998082/> (2018, January 2)

² Retrieved from: IBEF Indian Telecom Industry Analysis (2017, November 30)

³ Retrieved from: <http://money.cnn.com/2017/09/26/technology/india-mobile-congress-market-numbers/index.html> (2017, September 28)

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million⁴ had 4G enabled smart phones, placing the country third after China and USA in the global smart phones market. Jio was the first and only data-centric telecom operator in India based on VoLTE (Voice over Long Term Evolution) technology. This technology allowed the transfer of voice over IP (Internet Protocol) network, purportedly faster for voice as compared to a traditional network using 2G and 3G technology.

There had been immense growth in the telecom sector and the smartphone industry. The rate of data transfer in phones over the internet was rising exponentially (Exhibit 3). While the first generation (1G) did not make any provision for transfer of data, with 4G, customers obtained data speeds of more than 50mbps (with averages of more than 10mbps), with 5G on the horizon.⁵

On 4G, it was possible to convert voice into data packets and transfer over the net (VoIP). Its cost was less than traditional services. A major concern for Raj Pudipeddi (Director–Consumer Businesses, Airtel) was that VoIP calls (both national and international) were perceived as FREE by customers although there was a cost for internet service. Presently, more than 300 million people were using the internet through their smartphone (Exhibit 4), and this number was expected to double (600-650 million) by 2020.⁶ The new entrant Reliance Jio, an INR 200,000 crore startup, had propelled India to the leading position in mobile data consumption, i.e., 150 crore Gigabytes (GB) per month, surpassing the combined mobile data consumption of the USA and China, i.e., 130 crore GB. As the Indian Telecom Industry gravitated towards the Internet-based Technology, the Telecom Regulatory Authority of India (TRAI) had cut mobile call connection charges.⁷

The revenue, KPI's, as well as the financial parameters of all mobile operators declined sharply (Exhibits 5 & 6). The flourishing Indian mobility industry witnessed an annual revenue decline of 2% (approximately) in Financial Year (FY) 2017 for the first time in its history. As of Q3 2017 the Average Revenue per User (ARPU) of Airtel and Vodafone had tumbled to INR 145 and INR 146 per month from a year prior to INR 200 each, whereas Jio claimed an ARPU of INR 156 a month. Jio, BSNL and Airtel were the only telecom operators who added subscribers in September 2017 across India.⁸

This unprecedented disruption in the Indian wireless industry on free offers for both voice and data came from new entrant Jio who was well capitalised and aggressive. Also, the Goods and Services Tax (GST) was hiked from 15% to 18% and, TRAI reduced the interconnection usage charge (IUC) to 6 Paise from 14 Paise per minute in Q4 2017. This placed additional pressure on the incumbents. The cumulative debt of telecom players rose to INR 4.6 lac crore, driven in a

⁴ Retrieved from: <http://www.livemint.com/Industry/QYMhnlwLGztkZKZO4EEbK5L/India-to-overtake-US-as-secondlargest-4G-phone-base-Counte.html> (2017, July 28)

⁵ In mobile telecom there have been successive generations of telecom technology. The first major global standard was 2G (more widely known as GSM) in the 1990s. This was succeeded by 3G (also called UMTS) that was first launched in Japan in 1999 but became widespread only in the late 2000s globally. In India, it was introduced in 2004 by Airtel. 4G was introduced in 2012.

⁶ Retrieved from: IBEF news (2017, November 30)

⁷ Retrieved from: <https://economictimes.indiatimes.com/tech/internet/consolidations-done-telecom-looks-to-dial-up-growth-in-2018/articleshow/62314319.cms> (2017, December 31)

⁸ Retrieved from: <https://in.finance.yahoo.com/news/telecom-turmoil-bleeding-balance-sheets-054203539.html> (2017, December 18)

major part by spectrum licence payments, and the reduced revenues. The sector's adjusted gross revenue had fallen to INR 30,759 crore for the quarter ending, 2017, i.e., year-on-year decrease of 18.1%.⁹ A few sector consolidation activities had occurred, i.e., the merger of Vodafone India and Idea Cellular, Airtel's MoU with Tata Teleservices and TATA Teleservices Maharashtra, to merge their consumer mobile businesses, Reliance Communications completion of a merger with MTS and talks of a possible merger between BSNL and MTNL.

Overall, there was turmoil in the hyper-competitive telecom sector as the balance sheets of many major players were bleeding. The industry was struggling with a cumulative debt of INR 4.6 lac crores; the government was finalising a relief package for telecom operators to allow time for spectrum payments on the New Telecom Policy.

Company Background¹⁰

Bharti Airtel Limited "Airtel," the flagship company of Bharti Enterprises, had a diverse business portfolio and it had created global brands in the telecommunications sector. Started in 1995, Airtel was the leading mobile telecom operator in India with 285.95 million subscribers in India in Q3 2017.

In India, Airtel was the market leader, and its product offerings included 2G, 3G, 4G wireless services, fixed line services, mobile commerce, DTH, high-speed broadband and enterprise services that included national and international long distance service to carriers. In other geographies, it offered 2G, 3G, 4G wireless services and mobile commerce.

Sunil Bharti Mittal, the founder chairman of Bharti Enterprises, began his journey manufacturing spare parts for bicycles in the 1970's. His entrepreneurial instincts enabled quick sensing of new business opportunities. Bharti established itself as a supplier of basic telecom equipment. In the 1990's when the Indian government liberalised the sector and permitted private players to provide the telecom services, Bharti Enterprises availed the opportunity to become one of India's most respected brands and India's largest Telecommunications Company with a footprint in 22 telecom circles. Bharti Airtel's India business was structured into three individual strategic business units: Mobile Services, Airtel Telemedia Services and Enterprise Services.

Mobile Services provided mobile and fixed wireless services using Global System for Mobile (GSM) technology across the 22 telecom circles.¹¹ Airtel Telemedia services business offered telephone and broadband services in 95 cities and Direct to Home Service (DTH) and Airtel Digital TV. Enterprise services provided end-to-end¹² telecom solutions to corporate customers and national and international long distance services to carriers.

⁹ Retrieved from: <https://economictimes.indiatimes.com/news/company/corporate-trends/telecom-sector-in-2017-year-of-turmoil-with-bleeding-balance-sheets-under-gsts-shadow/articleshow/62115951.cms> (2017, December 22)

¹⁰ Retrieved from Airtel Website.

¹¹ A telecom service area based on population coverage.

¹² End to end solutions meant that the customer did not have to be concerned about any part of the telecom service, as it was addressed by the provider.

Bharti Airtel was the #1 service brand in India and had the vision to become the most loved brand.¹³ Management espoused a business model that was based on producing the lowest cost minutes (for end-consumers) while maintaining and growing margins. The motivation was that increased affordability (especially for end consumers) in a low-income country like India would drive more usage which would lead to scale economies and improved profitability; an idea that was threatened with the entry of Jio. In December 2017, Bharti Airtel had a wireless customer base of 290 million Indians and a revenue market share of 30.9%. Research also suggested that in 2014, of the data traffic on the internet, humans accounted for 44%, search engines and good bots accounted for 27%, and impersonators and hacking tools accounted for 25%.¹⁴

Globally, Bharti Airtel had operations in 17 countries across Asia and Africa. Headquartered in New Delhi, India, the company was ranked amongst the top three mobile service providers globally for the number of subscribers. It had a mobile subscriber base of more than 355.7 million and annual global revenues and profits in excess of INR 92,000 crores (USD 15 billion) and INR 5100 crores (USD 800 million) for FY 2017.¹⁵ Airtel was scouting for more data-based revenues from India (Exhibit 7) as subscriber and voice ARPUs¹⁶ growth had slowed, leading to a decline in net profits from INR 8900 crores in 2010-11 to losses in 2017-18 after the launch of Reliance Jio. Even SMS, a very high margin product was declining.

Competitors

Reliance Jio: Owned by Reliance Industries Ltd, Jio was the third largest telecom player in the sector. It aimed to replicate what China Mobile; China's state-owned mobile telephony provider did in recent years (starting 2014, in a matter of 18 months, China Mobile migrated 400 million users to LTE, at the same time adding 22-23 million subscribers in a month). Jio aimed to acquire 100 million customers in the first year of its operations and thereafter retain those customers whilst adding new ones. The company acquired 100 million customers in 170 days after its launch adding on average seven customers every second daily.

Jio triggered an unprecedented market disruption with its "One India" tariff plan with free voice and the lowest data rates in the world. This led to an intense price war in the 4G enabled network coverage infrastructure across all the 22 circles in India. Jio providing free calls to its subscribers made sustaining revenues from voice calls challenging for existing competitors. A study by Smartapp indicated that Facebook was the largest beneficiary of free data offered by Jio with a 467% increase in Facebook logins; the second biggest beneficiary was video apps such as YouTube, Hot Star, Amazon Prime and Netflix with a jump of 336% increase in usage.¹⁷

Idea Cellular¹⁸: India's first multinational corporation, Aditya Birla Group, owned Idea which was among the top three Indian mobile operators. It was a pan-India integrated wireless

¹³ Retrieved from: Airtel Management presentation May 2014.

¹⁴ Retrieved from: <http://www.statista.com/chart/1894/global-website-traffic-by-source/> (2015, April 21).

¹⁵ Airtel's Indian operations had annual revenues of INR 51,973 crores and profits of INR 10,713 crore in 2014-15. 1 crore = 10 million.

¹⁶ ARPU refers to Average revenue per user (sometimes known as Average revenue per unit) is the total revenue divided by the number of subscribers.

¹⁷ Retrieved from: <https://telecom.economictimes.indiatimes.com/news/reliance-jios-free-data-makes-big-contribution-to-facebook-revenue-growth-in-q4/56931784> (2017, December 29)

¹⁸ Retrieved from: Idea Cellular Annual Report 2017.

operator offering 2G, 3G and 4G services. Idea provided GSM-based mobile telecommunication services in all 22 telecom circles of India. Idea had 3G spectrums in 15 telecom circles and 4G spectrums in 20 telecom circles in India.

On 20 March 2017, Idea announced a merger between Vodafone India and Idea Cellular to create India's largest telecom operator. The combined company would be the leading communications provider in India with 400 million customers, 35% customer market share and 41% revenue market share. For the first time, Idea had reported revenue decline of 8% at INR 358,827 million for FY 2017 (INR 361625 million in FY 2016). The merger was completed on August 31, 2018.

Vodafone: Vodafone India was a 100% subsidiary of the Vodafone Group. It commenced its operations in 1994 when Hutchison Telecom acquired a cellular licence for Mumbai. The brand Vodafone was launched in 2007 after Vodafone Plc. acquired a majority stake in Hutchison Essar. The company was operating in 22 telecom circles.¹⁹ As at 2017, Vodafone India had maintained service revenue at INR 42956 crores, and data revenue was at INR 8467 crores (5% growth year-on-year). It had 66.9 million data users of which 43.5 million users consumed more than 1MB of data. The data ARPU was INR 140 (Q4 FY 2017).²⁰ As of August 31, 2018, Vodafone was part of Vodafone Idea, the merged entity; however, the two brands, Vodafone and Idea, continued to operate in the market.

BSNL: Bharat Sanchar Nigam Limited (BSNL), a large and leading public sector unit, provided a comprehensive range of telecom services in India. BSNL served its customers with various offerings namely Wireline, CDMA mobile, GSM mobile, Internet, Broadband, Carrier Service, VSAT, VoIP, etc. ²¹ BSNL's revenue from operations was INR 28477 crore in FY 2017.²²

Jio's Data Revolution and the Airtel Dilemma

With the entry of Reliance Jio in September 2016, a data revolution in India commenced. Initially, Jio offered free services until December 2016 which were extended till March 2017. This allowed Jio a six month period of "free" promotional offers.

As part of its entry, Infotel Broadband paid INR 12,847.77 crore (5,000 times its net worth of INR 2.5 crore) for the BWA spectrum. After winning the bid, it was acquired by Mr. Mukesh Ambani, Chairman and Managing Director of Reliance Industries Limited and renamed Reliance Jio. During Jio's launch, TRAI contacted DoT for clarification on the trial period of an operator. DoT responded in the first week of December 2016 by writing back to TRAI seeking its recommendations in this matter. Nine months after (1 May, 2017) DoT asked TRAI to look into the matter, TRAI floated a consultation paper on "Network Testing before Commercial Launch of Services". One of the questions in the paper was "Can a TSP be allowed to provide all the services free of cost to test users/subscribers even before commercial launch of services, is one of the issues that arose during the recent past. One view is that providing full-blown services during test phase may have implications on the IUC, pricing, quality of service

¹⁹ Retrieved from: Vodafone Website.

²⁰ Retrieved from: [http://www.communicationstoday.co.in/11665-vodafone-india-full-year-fy17-results\(2017,March 31\)](http://www.communicationstoday.co.in/11665-vodafone-india-full-year-fy17-results(2017,March 31))

²¹ Retrieved from: BSNL website.

²² Retrieved from: [https://telecom.economictimes.indiatimes.com/news/bsnl-sees-fy17-revenue-holding-up-despite-tough-competition/60113333\(2017, August 18\)](https://telecom.economictimes.indiatimes.com/news/bsnl-sees-fy17-revenue-holding-up-despite-tough-competition/60113333(2017, August 18))

(QoS) and other regulatory aspects and as such can be viewed against the spirit of a level playing field.”

Finally the recommendations were placed in the public domain on 4 December 2017 with the official advice that there ought be a limit of 90 days. It said, “There should be a limit of 90 days on the test phase involving test subscribers. However, if the TSP fails to conclude network testing due to valid reasons, it may make a representation to the Licensor, seeking additional time for network testing giving detailed justification, which may be decided by the Licensor on a case to case basis.”

As per the Telecommunications Tariff Order, 1990 the tariff plans need to be consistent with three crucial regulatory principles-non-discrimination, non-predation and interconnection usage charges (IUC) compliance.

Thereafter Jio launched Jio Prime and Jio Dhan Dhana Dhan (Exhibit 8). To push these offers to customers Jio dangled a 6.5% commission to retailers that was much higher than the commission offered by other telecom operators.

This action forced all competitors to slash their tariffs. Jio acquired 100 million customers in the first six months of its launch, with the aim to lead the market with 50% share by 2021 and eyeing Airtel's status as India's largest telecom provider. Jio's free services eroded the profitability of other operators (see Exhibit 5). Jio's market entry led to an intense price war, which resulted in rock-bottom LTE (Long Term Evolution) service offerings from all operators, driving more customers to 4G than ever before (Exhibit 9). Jio's entry with data as its primary source of revenue represented a shift in the structure of an Indian telecom operator as its competitors were earning a bulk of their services through voice revenues. Jio was the first to start 4G VoLTE service in India offering its services through LTE where data was transferred simultaneously with the voice call. It facilitated phone calls using IP (Internet Protocol). Airtel was the first of the established telecom operators to adopt the VoLTE technology after Jio.²³ Jio had gained the first mover advantage to transfer information via 4G. The other operators who were offering 4G service transmitted voice calls via 2G or 3G technological platforms for customers using VOIP.

Jio claimed to offer data to customers at a low price of INR 50 per GB, putting an end to voice call charges in India. Voice was free, for life (Exhibit 10). According to Jio, customers would now only pay for data. There would be no roaming charges and no charges for SMS. The incumbent operators notably did not react to Jio's entry. However, the price war started after Jio's promotion period ended (Exhibit 11).

Airtel slashed its 3G and 4G charges by 80% to as low as INR 51 per GB. Airtel also announced a special 4G data pack offering “free” data for 90 days. Vodafone revised its data prices to INR 100 for 300 MB, INR 250 for 1GB and so on. State-run BSNL also decided to match Jio's offering “tariff by tariff.” For its high usage customers, BSNL brought down its tariff to INR 1 per GB. This resulted in a decline of Adjusted Gross Revenue of the operators (Exhibit 12).

²³ Retrieved from: <https://telecom.economictimes.indiatimes.com/news/airtel-launches-volte-service-in-mumbai/60461982> (2017, September 12)

As per a study, 82% of Jio customers were using it as a secondary SIM, and the remaining 18% of customers as a standalone (Subscriber Identity Module) SIM. The study also stated that 86% of Jio subscribers were using the company services because of the “free” factor.²⁴ Given that Jio technology was enabled only for 4G and efficient use required that both the sender and the receiver use 4G LTE and VOLTE, when the other party did not have LTE or VOLTE, the efficiency and performance of the calls and data transfer rates declined substantially—the flip side of the network effect.

Airtel or Jio?

Before the entry of Jio, customers considered data and voice as separate. Jio blurred this mental segregation as it offered one price for both data and voice (where the latter was stated as free). This created a mix of curiosity, a desire to try and also scepticism in the minds of customers as they were unable to choose whether the free services offered by Jio were better or services by Airtel, who was in the market for longer, were superior. Exhibit 13 relays some tweets from customers on Airtel and Jio, with both having admirers and detractors. Many chose to retain Reliance as a second SIM and used it primarily for data; infact some of these consumers did not know their Reliance number. A screenshot from a twitter feed below indicates some usage and positioning issues for Jio. In general, network strength and coverage, call completion and the quality of call and the peer group using a network became more important for a customer as time passed.

Reliance Jio with its free services transformed the lives of different segments of society. It reduced the digital divide that existed between the rural and urban population and led to a fundamental behavior change with the average mobile internet user spending 70% time on social media and entertainment Apps.

Jio changed the way agrarian India used phones. It transformed the mode of communication into the mode for information by enabling video watching and information gathering and enhancing produce through better insights on products and service options available. It helped farmers shift from conventional modes to online platforms and to compete globally and get better returns.

The case of Sanju Devi, a 40-year-old handicrafts artist from Bhusara village in Bihar's Muzzafarpur District is illustrative. Sanju Devi started marketing her products by sharing images on social networking sites rather than downloading songs and videos.²⁵ Similarly, Roshan Setty, a fourth-generation small department store owner, in Malleshwaram, Bengaluru, started promoting offers relating to his business via WhatsApp. It also helped Laxmi Sharma, a villager from Alwar, Rajasthan, to take an informed decision about not buying a foreign breed of cow and saving her from the high maintenance cost of the cow.²⁶

²⁴ Retrieved from: <https://telecom.economictimes.indiatimes.com/news/only-18-reliance-jio-customers-use-connection-as-standalone-sim-82-use-it-as-second-velocity-mr-study/59219618> (2017, June 19)

²⁵ Retrieved from: <https://qz.com/755825/anatomy-of-a-rural-internet-user-in-india/> (2016, August 12)

²⁶ Retrieved from: <https://www.theguardian.com/global-development/2016/dec/05/india-women-mobile-internet-scheme-rural-digital-literacy> (2016, December 05)

A similar upswing was witnessed in urban India where videos accounted for 75% data consumption²⁷ powered by Hindi and regional language content, where 90% of the videos watched online consisted of Hindi and regional language videos. There were over 20 million avid video consumers who spent over 22 hours a month consuming video indicating a quantum increase in data usage.²⁸ While Reliance Jio had contributed to a step jump in data usage, it had its share of detractors and critics. A sample set of tweets reflecting this attitude is presented below.



²⁷ Retrieved from: <https://economictimes.indiatimes.com/tech/internet/fall-in-data-prices-new-users-reason-for-rise-in-video-consumption/articleshow/62289634.cms>

²⁸ Retrieved from: <https://economictimes.indiatimes.com/tech/internet/average-mobile-data-usage-at-11gb-a-month-nokia/articleshow/63032695.cms>

Voice and data calling in India were primarily priced on a pre-paid basis with 93% of connections being pre-paid with no contract. A typical pre-paid voice plan was a “recharge” of INR 105 (USD 1.6) that would give local voice minutes at INR 0.5 per minute, and long distance (inter circle) calls at INR 1 per minute (1.35 USD Cents per minute). Of the INR 105 paid, typically, about INR 15-20 would be a standing charge (the price of call minutes times price per minute would add up to only about INR 90). This would be valid for up to 6 months regardless of actual voice minutes used. A typical data plan was for 21 days and 300Mb of data usage for INR 105.

Postpaid contracts had higher standing charges with a contract. (Unlike the USA, however, the contract did not include a subsidised handset tied to an operator). A typical postpaid contract with a higher end user was for INR 299 per month with 300 free local calls, and long distance calls at INR 0.50 per minute. In addition, the user had a 3G data pack at INR 750 per month for unlimited data. In June 2014 data usage by mobile subscribers was estimated at 70 MB per subscriber per month.²⁹

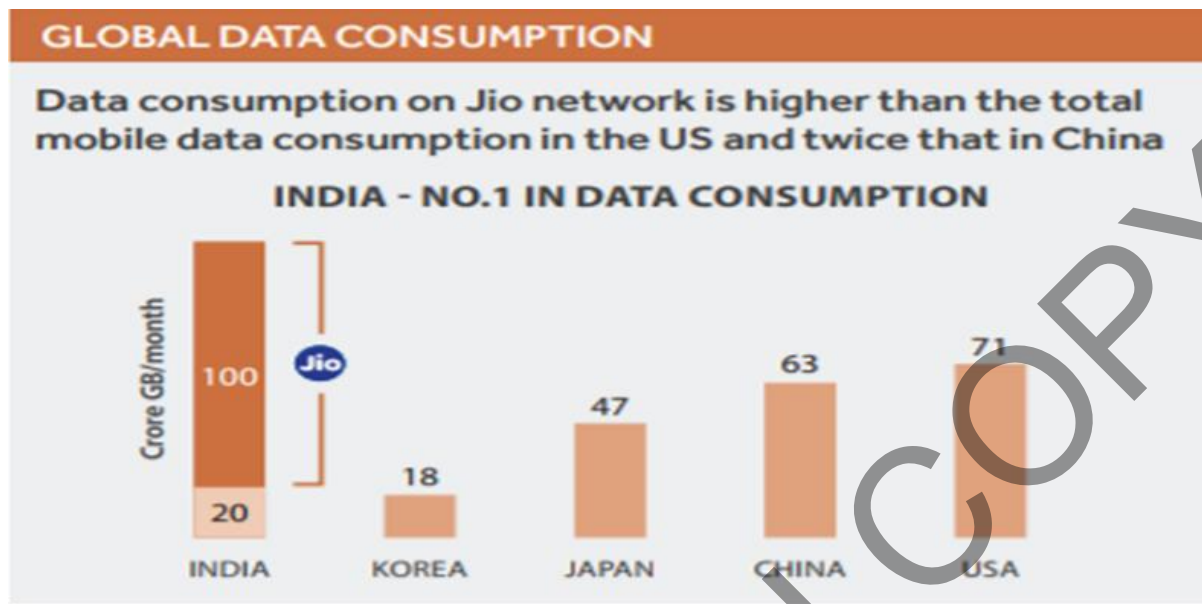
Looking Ahead: What Next?

Vittal wondered “What would be an effective pricing strategy and approach to counter Jio and mitigate the effects of cannibalisation?” He had to find a way to hold on to his customer base as well. An associated challenge he pondered “..the time it would take the industry to transition from its present hybrid state to one that was entirely internet based” Vittal and Pudipeddi needed a point of view that they could present to the Board.

²⁹ Retrieved from:

<http://www.trai.gov.in/WriteReadData/WhatsNew/Documents/quarterly%20press%20release%20-final.pdf>
(2015, April 21).

Exhibit 1: Data and Voice consumption pattern in India



Source: Reliance Industries Limited Annual Report 2016-17

	Q3 2016	Q4 2016	Q1 2017	Q2 2017	Q3 2017
Data Consumption (billion MB)	763	2939	3546	4307	5560
Voice Consumption (billion minutes)	1137	1170	1390	1508	1545
Data Usage (MB/Subscriber/Month)	772	2806	3144	3628	4638
Voice Usage (Minutes/Subscriber/Month)	363	358	403	427	435

Source: <http://www.financialexpress.com/industry/data-revenue-of-telecom-firms-jumps-by-staggering-59-pct-to-rs-11900-cr-in-q2/998082/> (2018, January 2)

Exhibit 2: Telecom Subscribers in India (Million)

	Q3 2017	Q2 2017	Q1 2017	Q4 2016	Q3 2016	Q2 2016	Q1 2016	Q4 2015	Q3 2015	Q2 2015	Q1 2015
Total Subscribers (Wireless +Wireline)	1206.7 1	1,210.8 4	1194.5 8	1151.7 8	1074.2 4	1059.8 6	1058.8 6	1,036. 41	1022.6 1	1006.9 6	996.49
Total Wireless Subscribers	1183.0 4	1,186.8 4	1170.1 8	1127.3 7	1049.7 4	1035.1 2	1033.6 3	1,010. 89	996.66	980.81	969.89
Total Wireline Subscribers	23.67	24	24.4	24.4	24.49	24.74	25.22	25.52	25.95	26.15	26.59
Teledensity	91.56	93.98	92.98	89.9	84.09	83.2	83.36	81.83	80.98	79.98	79.38
Total Internet Subscribers	429.23	431.21	422.19	391.5	367.48	350.48	342.65	331.66	324.95	319.42	302.35
Wired Internet Subscribers	21.35	21.67	21.58	21.51	21.26	20.76	20.44	19.98	19.6	19.21	19.07
Wireless Internet Subscribers	407.88	409.55	400.62	370	346.22	329.72	322.21	311.69	305.35	300.22	283.29
Total Outgoing MOU for Internet Telephony	283	260	258	251	267	271	277	271	294	272	245

Telecom Operator Wise Subscribers (Million)

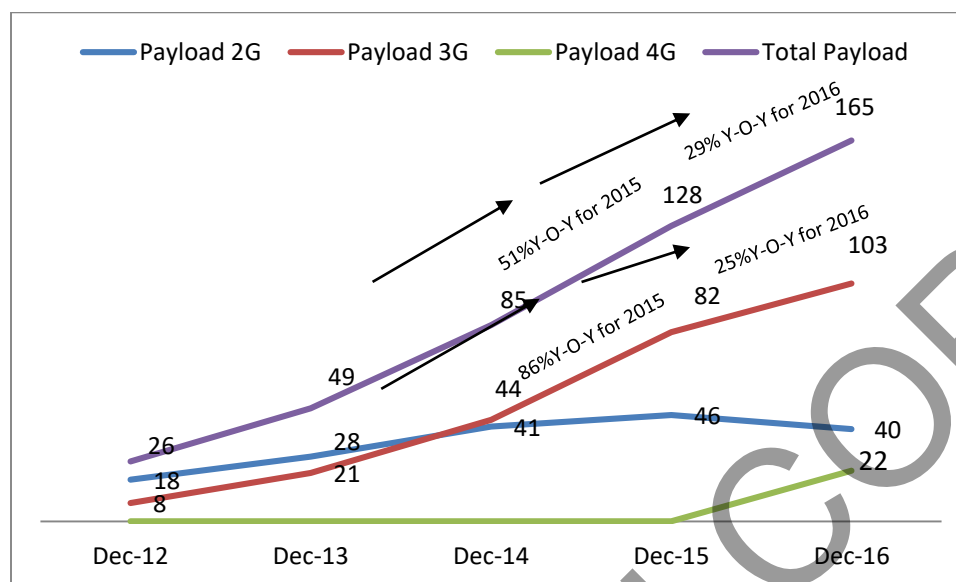
	Q3 2017	Q2 2017	Q1 2017	Q4 2016	Q3 2016	Q2 2016	Q1 2016	Q4 2015	Q3 2015	Q2 2015	Q1 2015
Bharti Airtel	285.95	284.53	277.51	269.68	263.74	259.47	254.9	246.89	238.74	234.11	229.43
Vodafone	207.63	212.09	209.2	204.82	200.84	199.49	198.04	193.69	188.26	185.47	183.88
Idea	190.16	196.28	195.37	190.52	178.82	176.23	175.07	171.91	166.56	162.08	157.81
BSNL	118.61	117.37	114.68	110.53	107.66	103.74	101.11	97.64	95.34	93.29	93.64
Reliance Jio	138.62	123.36	108.68	72.16	15.98						
Tata	46.76	45.54	50.74	54.71	58.79	61.1	61.82	62.43	63.73	63.26	67.99

Telecom Operator Wise Internet Subscribers (Million)

	Q3 2017	Q2 2017	Q1 2017	Q4 2016	Q3 2016	Q2 2016	Q1 2016	Q4 2015	Q3 2015	Q2 2015	Q1 2015
Bharti Airtel	94.3	94.32	87.4	84.83	95.31	92.52	90.64	84.07	82.86	82.6	76.31
Vodafone	67.71	69.16	66.86	65.05	69.6	69.72	67.55	67.45	66.51	66.83	63.75
Idea	38.2	38.13	42.25	48.6	54.08	49.07	44.04	41.37	41.35	37.16	33.43
BSNL	32.15	33.91	35.75	32.9	34.69	34.35	34.09	33.37	32.87	32.52	32.02
Reliance Jio	138.61	123.36	108.68	72.16	16						
Tata	9	11.13	13.31	14.77	17.95	19.89	21.04	20.9	20.64	20.34	20.49

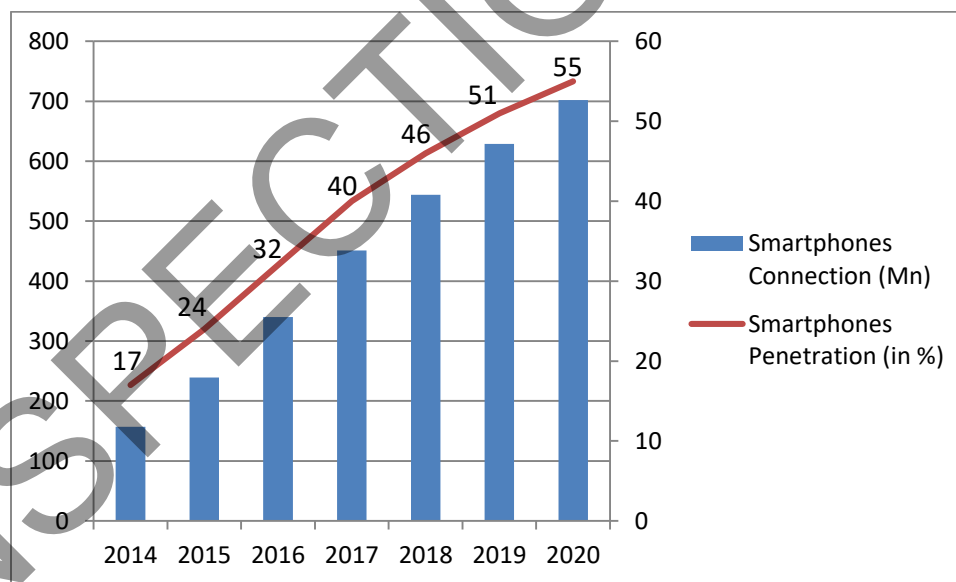
Source: TRAI Performance Indicator Reports

Exhibit 3: Pan-India mobile data usage (in Petabytes (PB))



Source: India Mobile Broadband Index 2017

Exhibit 4: Smartphone adoption trends in India



Penetration is the number of active smartphone mobile phone users per 100 people within a specific population.

Source: COAI Annual Report 2016-17

Exhibit 5: Comparative Financial Statements

	Bharti Airtel Ltd.					Reliance Jio Infocomm Ltd.				
(INR Million)	Sept'16	Dec'16	March'17	June'17	Sept'17	Sept'16	Dec'16	March'17	June'17	Sept'17
Total Expenses	1,49,857.00	1,48,679.00	3,16,403.00	1,44,828.00	1,39,954.00	41.9	54.3	176.1	215.7	64,193.20
Operating Expenses	98,527.00	95,483.00	93,198.00	95,977.00	92,173.00	47.6	66.7	252.8	314.6	47,052.80
Network Cost	35,789.00	37,151.00	37,262.00	36,802.00	35,191.00					13,718.90
Regulatory Charges(Licence/spectrum)	18,320.00	16,626.00	15,572.00	15,408.00	14,632.00	2.7	1.9	1.6	2.3	3,990.00
Access Charges	22,881.00	18,967.00	18,414.00	21,001.00	21,985.00					21,398.80
PBT	30,685.00	9,797.00	-1,32,482.00	4,223.00	5,358.00	-59.6	-78.8	-264.6	-325.2	-4,138.10
PAT	24,012.00	4,218.00	-1,41,762.00	2,806.00	4,945.00	-39	-51.5	-173.4	-212.7	-2,705.90
Marketing Expenses	7,951.00	8,131.00	8,033.00	7,850.00	6,578.00	1.7	6.1	167.5	209.6	2,608.40
Revenue	1,64,326.00	1,52,042.00	1,42,998.00	1,45,065.00	1,39,801.00					61,470.60

Source: Company Annual Reports

Exhibit 6: Revenue Trend

	Revenue Market Share (in %)				
Company	Q1 2016	Q2 2016	Q3 2016	Q4 2016	Q1 2017
Bharti Airtel	30.6	31.4	31.5	31.8	31.7
Vodafone India	23	23	22.6	22.4	22.7
Idea Cellular	18.8	19.1	19	19.5	20.2

Source: http://www.business-standard.com/article/companies/bharti-airtel-vodafone-idea-control-74-of-market-116061301029_1.html, (2016, June 14)

Exhibit 7: Airtel and Jio: Voice & Data Usage

	Q3 2017	Q2 2017	Q1 2017	Q4 2016	Q3 2016	Q2 2016	Q1 2016	Q4 2015	Q3 2015	Q2 2015	Q1 2015
Customer Base ('000)	282047	280647	273648	265853	259941	255735	251237	243289	235212	230662	226017
Voice Minutes on Network (Mn)	437142	421912	381236	330217	313403	314831	307988	290459	282138	290802	277869
Voice usage per customer (min)	518	507	471	419	406	414	415	405	404	424	418
Data Customer Base ('000)	65769	62560	57362	54915	62659	58903	58216	54860	51013	49470	46386
Data Customers as % of customer base	23.30	22.30	21	20.70	24.10	23	23.20	22.50	21.70	21.40	20.50
Data Usage per customer (MBs)	4087	2611	1331	972	1000	904	859	843	765	706	656
Total MB's on Network (Mn MBs)	7,83,809	4,72,385	2,25,109	1,71,817	1,78,125	1,58,035	1,46,768	1,33,946	1,14,960	1,02,015	86,627

Reliance Jio		
	Q4 2017	Q3 2017
Customer Base (mn)	160.1	138.6
Voice Minutes on Network (crore min)	31,113	24030
Voice usage per customer (min)	694	625
Data Usage per customer (GB)	9.6	9.6
Total MB's on Network (Crore GB)	431	378
VoLTE voice per day (crore minutes)	338	267

Source: Airtel and Jio Quarterly Reports

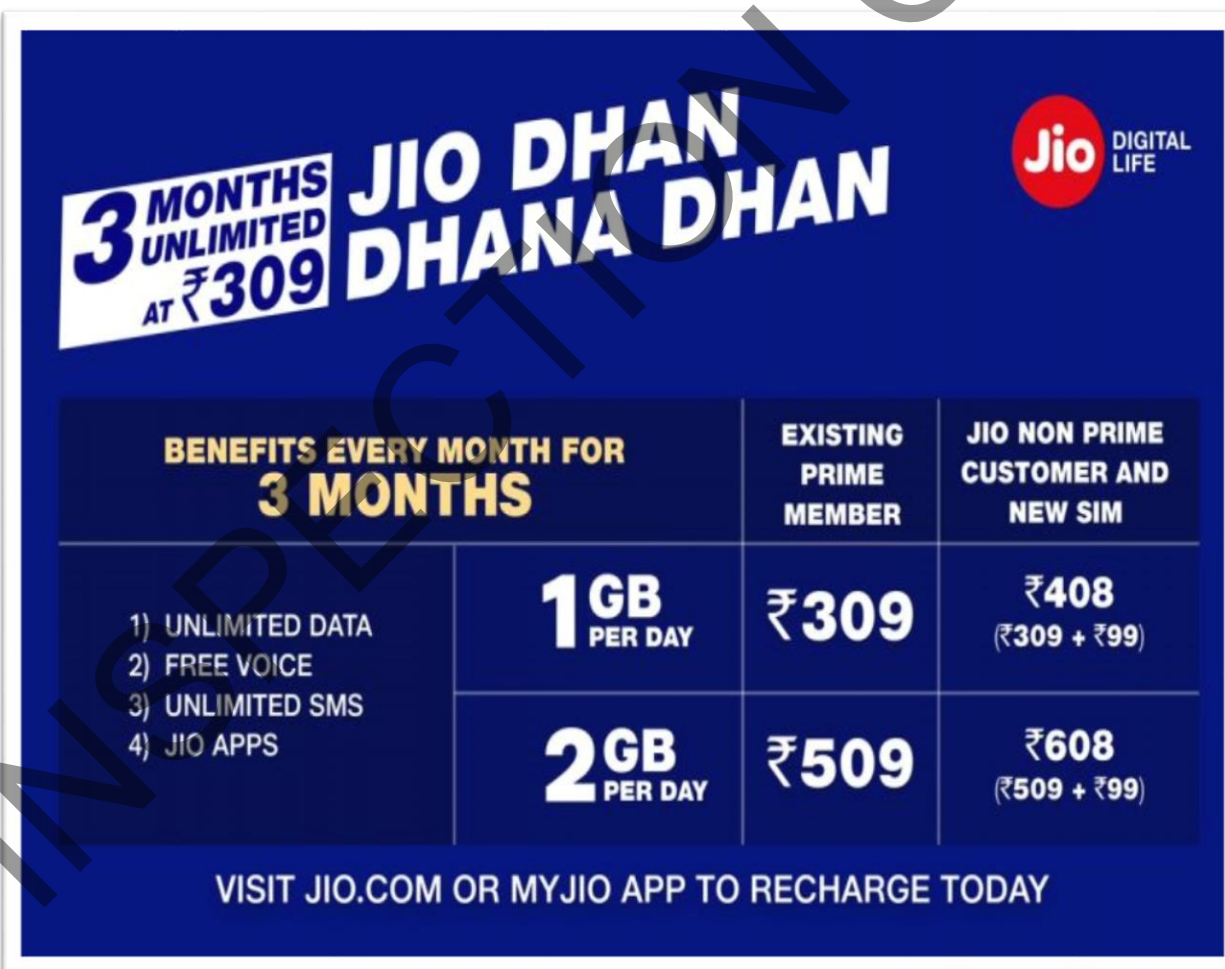
Exhibit 8: Jio Advertisements



JOIN JIO PRIME
ONE-TIME MEMBERSHIP AT ₹99
₹303/MONTH

UNLIMITED 4G DATA AT ₹10/DAY* | FREE VOICE CALLS FOR LIFE (LOCAL/STD/ROAMING) | NOW HOME-DELIVERED IN YOUR CITY

Source: Reliance Jio Twitter Handle (2017, March 27)



3 MONTHS UNLIMITED AT ₹309 **JIO DHAN DHANA DHAN** **Jio DIGITAL LIFE**

BENEFITS EVERY MONTH FOR 3 MONTHS		EXISTING PRIME MEMBER	JIO NON PRIME CUSTOMER AND NEW SIM
1) UNLIMITED DATA 2) FREE VOICE 3) UNLIMITED SMS 4) JIO APPS	1GB PER DAY	₹309	₹408 (₹309 + ₹99)
	2GB PER DAY	₹509	₹608 (₹509 + ₹99)

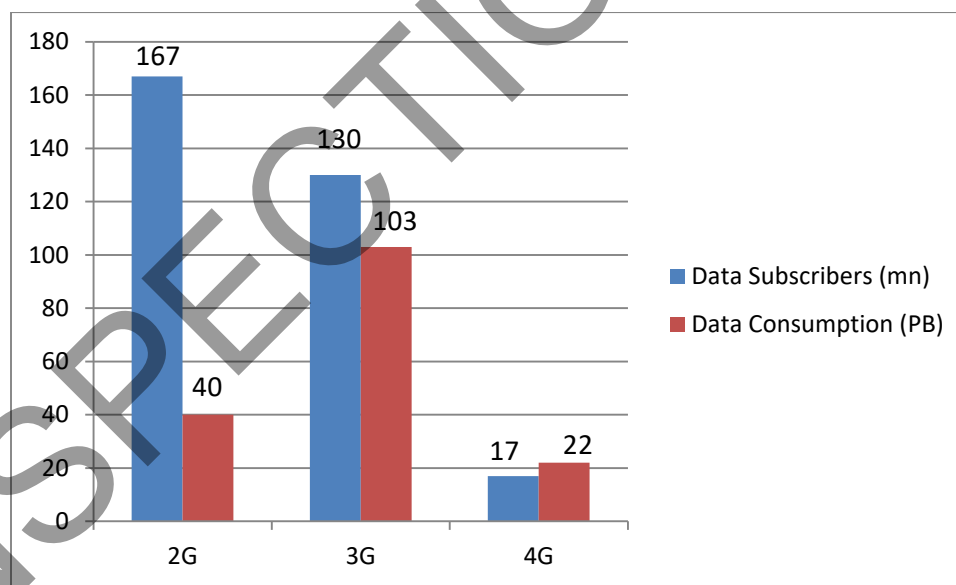
VISIT JIO.COM OR MYJIO APP TO RECHARGE TODAY

Source: <https://www.deccanchronicle.com/business/companies/110417/finally-reliance-jio-launches-dhan-dhana-dhan-offer-for-rs-309.html> (2017, April 11)



Source: <https://www.datareign.com/reliance-jio-happy-new-year-offer.html> (2017)

Exhibit 9: Mobile Data traffic in India



Source: Nokia MBIT Report 2017

Exhibit 10: Jio Recharge Plans unveiled during its commercial launch

	S	M	M	L	XL	XXL	XXXL
Jio Tariffs (INR)	149	499	999	1499	2499	3999	4999
FREE VOICE Unlimited Local STD Roaming to any network in India	✓	✓	✓	✓	✓	✓	✓
4G DATA	.3GB	4GB + Unlimited 4G at Night	10GB + Unlimited 4G at Night	20GB + Unlimited 4G at Night	35GB + Unlimited 4G at Night	60GB + Unlimited 4G at Night	75GB + Unlimited 4G at Night
Jio Apps Subscriptions worth INR 1,250	✓	✓	✓	✓	✓	✓	✓
Unlimited SMS*(L+N)	100	✓	✓	✓	✓	✓	✓
WiFi (JioNet Hotspot)	-	8GB	20GB	40GB	70GB	120GB	150GB
Validity of Pack (Days)	28	28	28	28	28	28	28

Source: <http://www.bgr.in/news/reliance-jio-4g-tariff-plans-announced-heres-everything-you-need-to-know/> (2016, September 1)

Exhibit 11: Jio vs. Airtel vs. Vodafone vs. Idea Recharge Plans

As the new entrant Reliance Jio entered the Indian Telecom Industry, there wasn't any notable difference in the plans of incumbent players.

	Airtel Rs. 145	Airtel Rs. 345	Vodafone Rs. 144	Vodafone Rs. 344	Idea Rs. 148	Idea Rs. 348	BSNL Rs. 144
Validity	28 days	28 days	28 days	28 days	28 days	28 days	28 days
Data	300 MB 4G data	1 GB 4G data	300 MB 4G data	1 GB 4G data	300 MB 4G data	3 GB 4G data	300 MB 3G data
Calls	Unlimited to Airtel numbers	Unlimited to any network	Unlimited to Vodafone numbers	Unlimited to any network	Unlimited to Idea numbers	Unlimited to any network	Unlimited to any network
Price (INR)	145	345	144	344	148	348	144

Source: <https://hub.91mobiles.com/reliance-jio-vs-airtel-vs-vodafone-vs-idea-unlimited-voice-calls/> (2017, February 3)

For Q2 2017: As the new entrant Jio started acquiring customers at a rapid pace through its low-priced data plans, the incumbent operators countered Jio by matching the prices and plans offered by Jio.

	Jio 303	Airtel 345	Vodafone 346	Idea 348	BSNL 339
Validity	28 Days	28 Days	28 Days	28 Days	28 Days
Recharge Type	Jio prime plan for all users	Segmented Offer. Initially to be rolled out for everyone	Segmented Offer	Segmented offer	For all BSNL users
Voice- Local, STD, On-net calling, off-net calling	Unlimited	Unlimited	Unlimited with FUP (3000 minutes per day/12000 minutes per week)	Unlimited	Unlimited BSNL to BSNL calls
Data	28 GB- 1 GB per day - After hitting the daily limit, speed will reduce to 128 kbps	28 GB-.5 GB in day and .5 GB at night (12 AM- 6 AM)	1 GB per day (28 Gb a month) and 4p/10 kb after hitting the limit	500 MB per day and 4p/10 kb after hitting the limit	2 GB per day (56 GB a month)
Additional Benefits	5 GB on all recharges before April 15, 2017	N/A	Double data & validity one time on first-time pay out of 346 before March 15, 2017	N/A	N/A
Plan Cost (INR)	303	345	346	348	339

FUP-Fair Use Policy (The speed is rationalised, i.e., reduced from normal use of the plan, on reaching fair usage level.)

Source: <https://telecom.economictimes.indiatimes.com/news/comparison-data-voice-offers-from-reliance-jio-airtel-vodafone-idea-and-bsnl/57955606> (2017, April 2)

For Q4 2017: During this phase, the new entrant started raising its prices selectively.

Telcos	Benefits						
	Data	Free Local+STD	Free SMS	Roaming	Apps Offered	Plan Cost (INR)	Validity
Reliance Jio	1GB/day	Unlimited	Unlimited	Unlimited	Jio Apps	459	84 days
Airtel	1GB/day	Unlimited	100 SMS/day	Free outgoing on national roaming	Airtel TV, Unlimited Wynk Music	448	70 days
Vodafone	1GB/day	Up to 250 mins per day (1000 mins a week)	100 SMS/day	Up to 250 mins per day (1000 mins a week)	Vodafone Pay Subscription	458	70 days
Idea	1GB/day	Unlimited	100 SMS/day	Free outgoing on national roaming	NA	498	70 days
BSNL	1GB/day	50 paise per min	NA	35 paise/min	NA	459	90 days

Source: <https://gadgetstouse.com/featured/best-1gb-per-day-plans-jio-airtel/90265> (2017, November 15)

Exhibit 12: Adjusted Gross Revenue of Telecom operators

Adjusted Gross Revenue (in Millions)										
	Q3 2016	% Change	Q4 2016	% Change	Q1 2017	% Change	Q2 2017	% Change	Q3 2017	% Change
Bharti Airtel	124931.8	-2.84	114956.6	-7.98	103618.8	-9.86	98882.9	-4.57	88466.3	-10.53
Vodafone	87547.8	-2.77	83044.9	-5.14	72581.3	-12.6	70894.2	-2.32	63046	-11.07
Idea	73178.1	-4.56	69579.5	-4.92	64011.3	-8	61609.1	-3.75	53078.1	-13.85
BSNL	32835.7	4.84	31461.1	-4.19	33187.1	5.49	30127.3	-9.22	24319.1	-19.28
Reliance Jio	234.1	183.81	-3366.5	-1537.8	-11786.1	250.1	-10365.6	-12.05	42735.9	512.29
Tata	25313.4	-3.4	22093.8	-12.27	18773.5	-15.03	17322.6	-7.33	16654.5	-3.86

Source: Quarterly Reports

Exhibit 13: Tweets of customers

